Mihailo Radović

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Research interests: flow matching and diffusion, JEPA world models, and robot learning, with the long-term goal of general-purpose autonomous agents.

Education

University of Belgrade, School of Electrical Engineering — Belgrade, Serbia
MSc in Electrical Engineering and Computing, Applied Mathematics module *GPA: 9.8/10* — *2025 – present*

Union University, School of Computing (RAF) — Belgrade, Serbia
BSc in Computer Science *GPA: 8.72/10* — *2021 – 2025*

Thesis: *SpriteFlow: Flow matching for pixel-art character generation*, advised by Prof. Nemanja Ilić.

Publications

Flow Matching Policy for Behavioral Cloning [DOI] [PDF] [code]
M. Radović, F. Marčić. *IcETRAN 2026*, IEEE.
Models the behavioral cloning action distribution as a continuous-time flow; competitive with or better than Gaussian and diffusion policies on D4RL benchmarks (MuJoCo locomotion, Franka Kitchen, AntMaze).

Presentations & Summer Schools

Eastern European Machine Learning Summer School (EEML 2026) — Cetinje, Montenegro
Selected participant; poster: Flow Matching Policy for Behavioral Cloning — *Jul – Aug 2026*

IcETRAN 2026
Paper presentation: Flow Matching Policy for Behavioral Cloning — *2026*

Projects

SpriteFlow: Flow Matching for Pixel-Art Character Generation (BSc thesis) [code]
Flow matching model with a time-conditioned U-Net generating 128×128 RGBA sprites; ODE sampling, linear and cosine noise schedules, EMA, and FID-based evaluation and model selection.

Policy Learning Baselines: PPO from Scratch [PyTorch] [JAX]
Reimplemented PPO from the original paper and compared it with A2C and vanilla policy gradient on MuJoCo locomotion (HalfCheetah, Hopper, Walker2d, Swimmer; 3 seeds each). Wrote a separate JAX implementation with a redesigned reward signal, trained on MuJoCo Playground (MJX) Cartpole Balance and Swing-Up.

Football Analysis [code]
Open-source video analysis tool with 180+ GitHub stars: YOLO11 detection, ByteTrack tracking, keypoint-based homography for player positions, and K-means team assignment from jersey colors.

Writing

Technical Blog mihailoradovic.com/blog
Write-ups on my research and the mathematics behind it, e.g. *Learnings on f-divergences* (hypothesis testing, inequalities, Wasserstein distance as an alternative, and their role in GANs and flow matching).

Honors & Awards

First Place, Google Build with AI Hackathon May 2025 [post]
SafeSpace: multi-agent hate-speech detection platform with a Gemini detector fine-tuned on Serbian data.

Skills

Programming: Python, C, C++, CUDA, SQL

ML & tools: PyTorch, JAX, Hugging Face Transformers, Gymnasium, MuJoCo / MJX, scikit-learn, OpenCV, NumPy, pandas, Weights & Biases, Git, LaTeX

Experience

Nordeus — Belgrade, Serbia
Junior Data Engineer — *2025 – present*